

# Alireza Laalmanesh

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## Education

### University of Waterloo

Sep. 2025 – Present

*Master of Mathematics in Computational Mathematics (Co-op)*

*Waterloo, Canada*

- Awarded the i-Comp-Data Fellowship (Inter-faculty award for Computation & Data research).
- Advisors: Dr. David Del Rey Fernández & Dr. Leia Minaker.
- Research Areas: Numerical Analysis, Deep Learning applications in Urban Data Science, Discrete Event Simulation.

### Amirkabir University of Technology (Tehran Polytechnic)

2019 – 2024

*B.Sc. in Industrial Engineering & Information Systems*

*Tehran, Iran*

- Cumulative GPA: 3.5/4.0.
- B.Sc. Thesis: "Operating Room Scheduling utilizing Meta-heuristic Optimization Algorithms".

## Technical Skills

**Languages:** Python, C/C++, Bash Scripting

**AI & Data Science:** PyTorch, scikit-learn, CNN Architectures (EfficientNet), DeepInsight, XGBoost

**Mathematical Context:** Linear Algebra, Numerical Optimization, Loss Function Implementation, Stochastic Modeling

## Professional Experience

### University of Waterloo

Sep. 2025 – Present

*Graduate Teaching Assistant (Faculty of Mathematics)*

*Waterloo, Canada*

- Courses: Calculus 2 for Sciences, Applied Linear Algebra, Algebra for Honours, Linear Algebra for Engineering.
- Conducted weekly tutorials and office hours to support student learning.
- Responsible for grading assignments and examinations.

### TAV Institute

2024

*Intern - Layout Design*

*Tehran, Iran*

- Contributed to factory layout analysis and process flow modeling.
- Assisted in capacity analysis regarding material handling systems.

## Selected Projects & Research

### Academic Hub Automation | *Linux Integration*

- Implementation of a local system to bridge Linux environments with Microsoft 365 services.
- Uses email transport layers for notification and calendar synchronization.

### Server Config Utility | *Bash, Linux Admin*

- Bash script designed for the initialization of headless Linux servers.
- Automates Tailscale and XRDP configuration for remote access setup.

### CrashVision-DL: Traffic Severity Prediction | *PyTorch, DeepInsight*

- Pipeline construction for converting tabular data (7.7M records) into feature images using t-SNE.
- Utilization of a custom Soft F-Beta loss function ( $\beta = 2.0$ ) to address recall in fatal crash detection.

### Android Hardware Bridge | *Linux, USB/IP*

- Configuration of Android devices as USB servers using USB/IP protocols.
- Involves binary injection via ADB to expose hardware interfaces over TCP/IP.

### CodeBase Aggregator | *Python, File I/O*

- Utility script for aggregating code repositories into structured datasets.
- Facilitates code analysis and context handling for Large Language Models (LLMs).

### Scientific Content Pipeline | *Gemini Vision, Python*

- ETL pipeline using PyMuPDF and Gemini Vision for converting scientific documents into audio notes.
- Features non-blocking architecture for text parsing and audio generation.

## Honors & Awards

**i-Comp-Data Fellowship:** Inter-faculty award for research in Computation & Data.

**Ranked in Top 1%** in National University Entrance Exam (among approx. 150,000 candidates).

Merit-based Scholarship recipient at Amirkabir University of Technology.